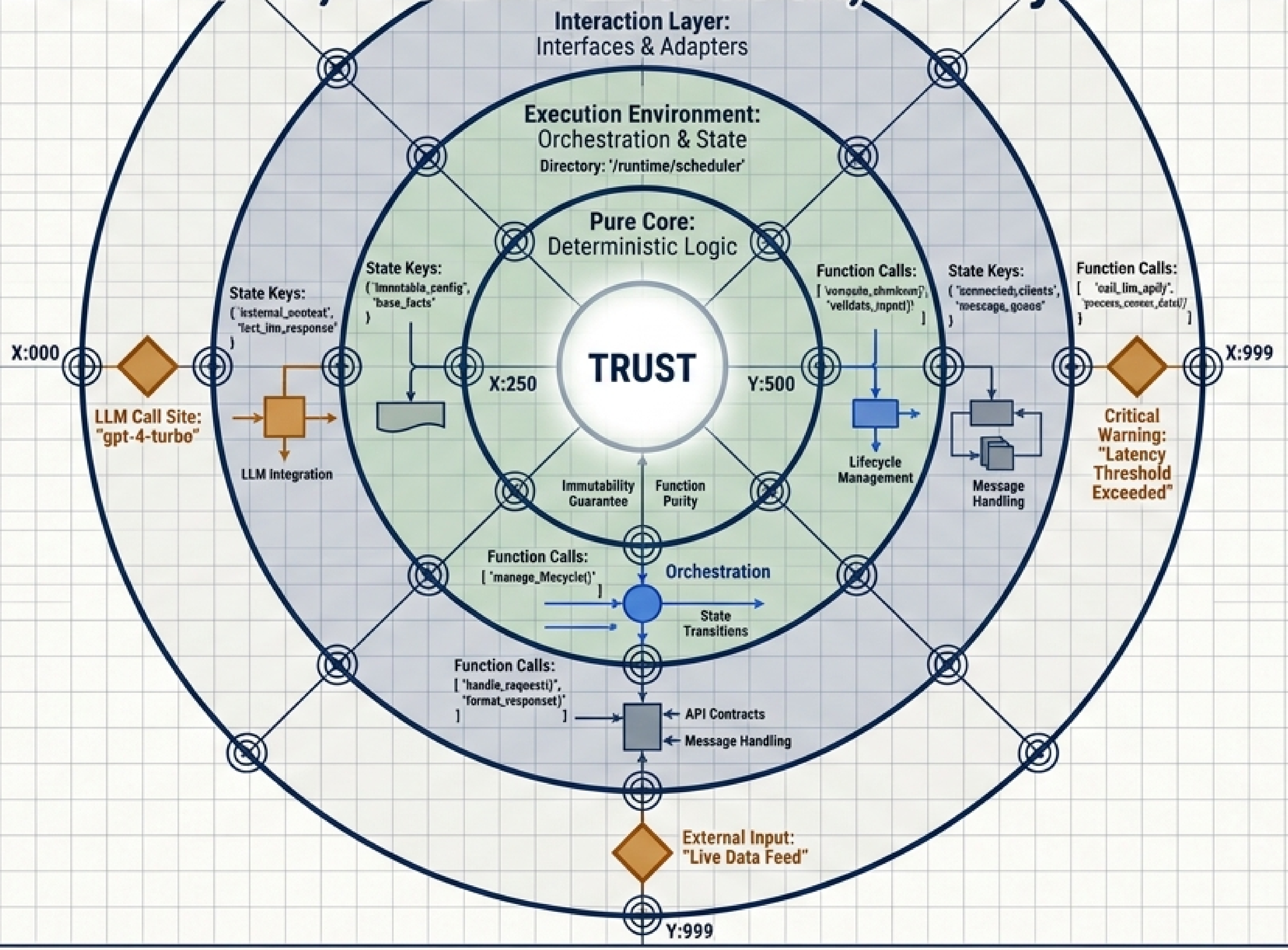
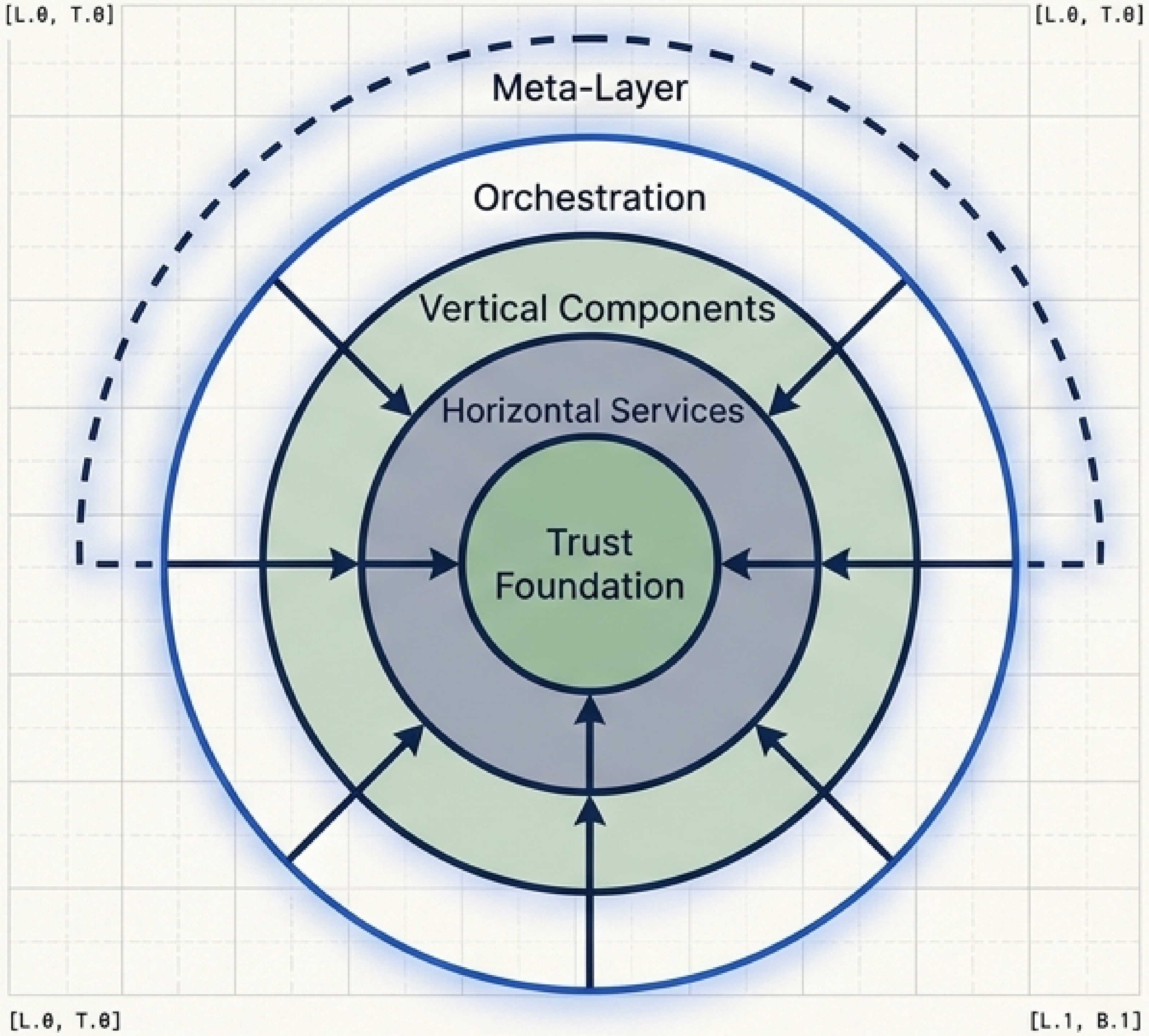
The Blueprint of an Autonomous Agent System

Structural Constraints, Runtime Execution, and Systemic Guarantees

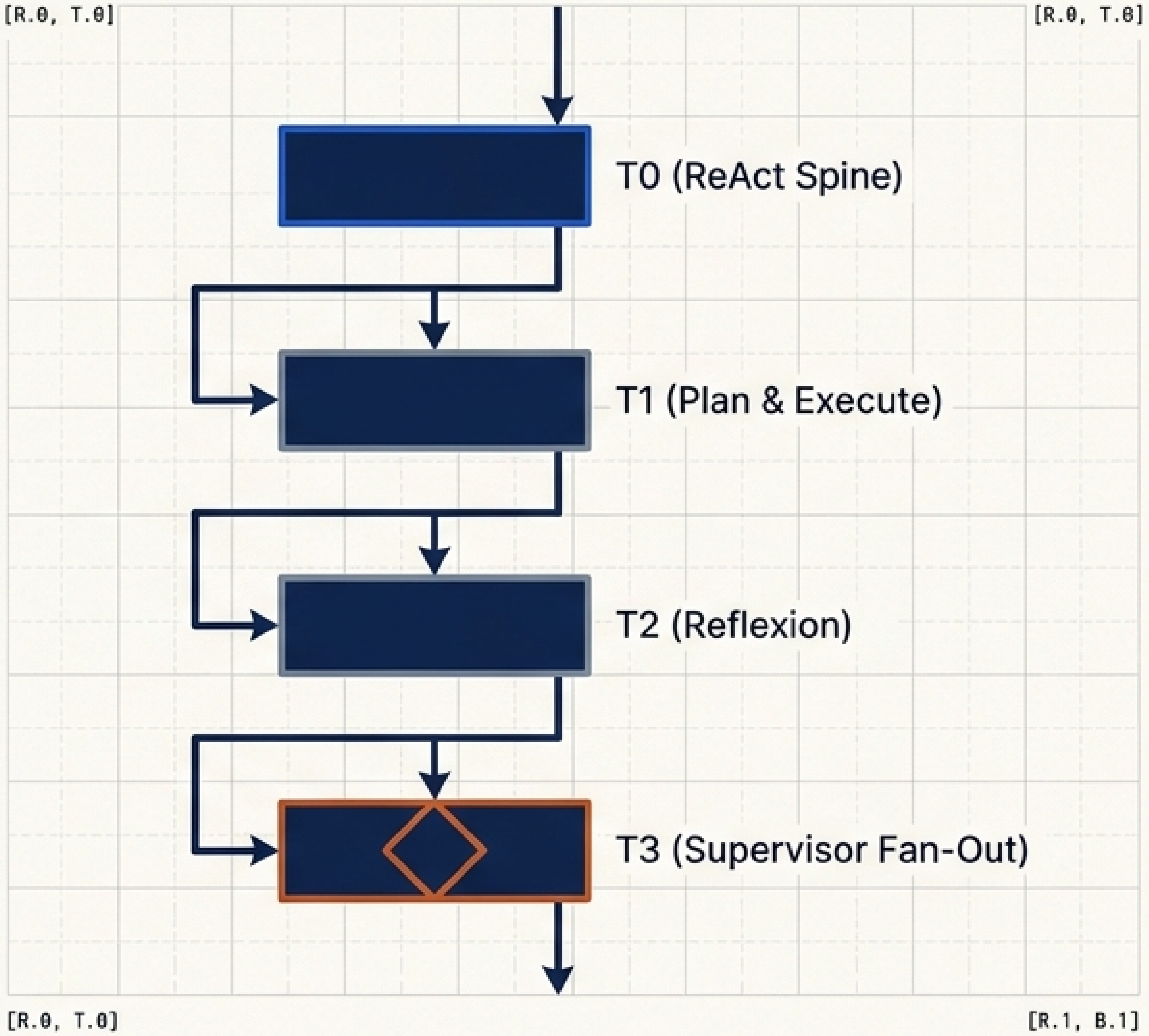


Two orthogonal axes define to the agent's behavior



The Structural Constraints

Strict dependency rules dictate where code lives and what it can import. Dependencies only point inward.



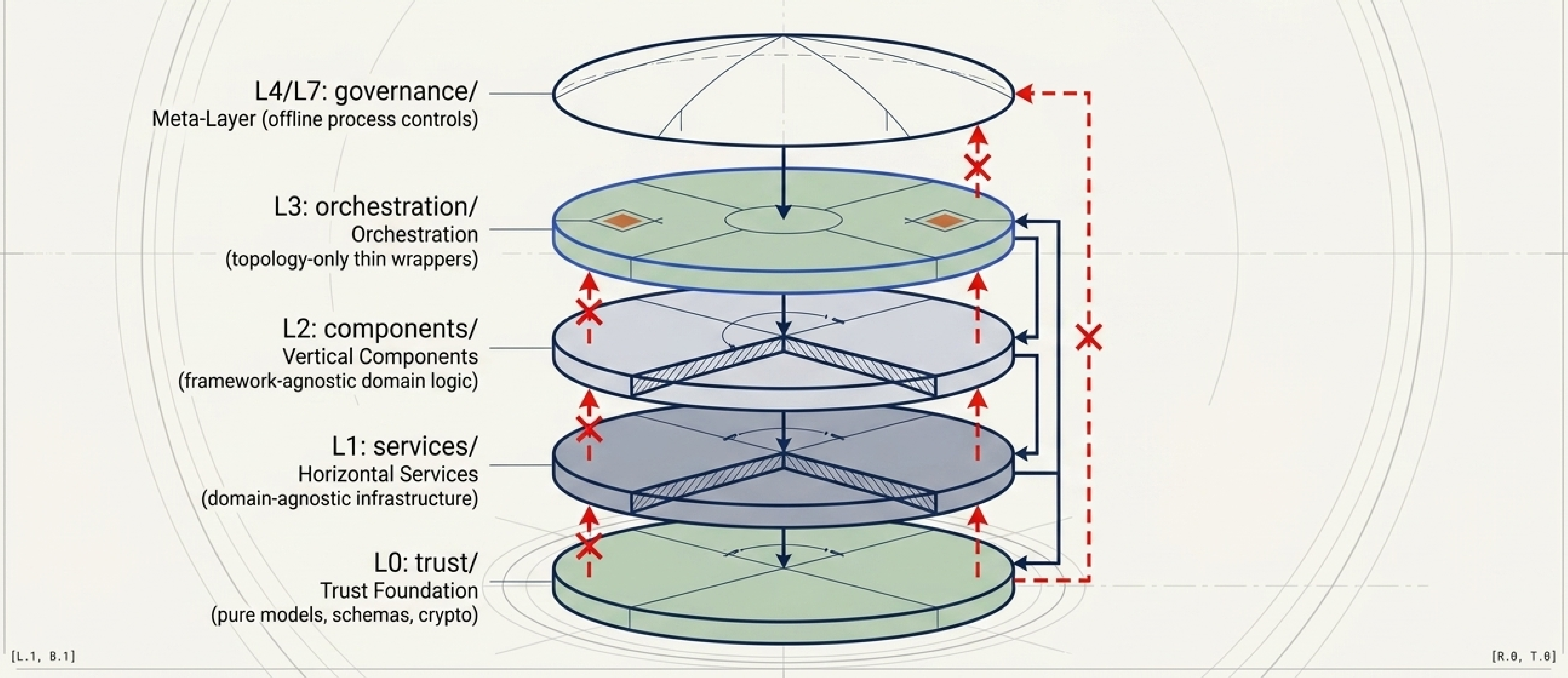
The Runtime Execution

Orthogonal mechanisms execute the logic, combining planning depth with additive pipeline tiers.

NotebookLM

The 4+1 Layer Architecture enforces a unidirectional dependency hierarchy

No upward imports. The Trust Foundation acts as the domain core (Shared Kernel), entirely dependency-free.



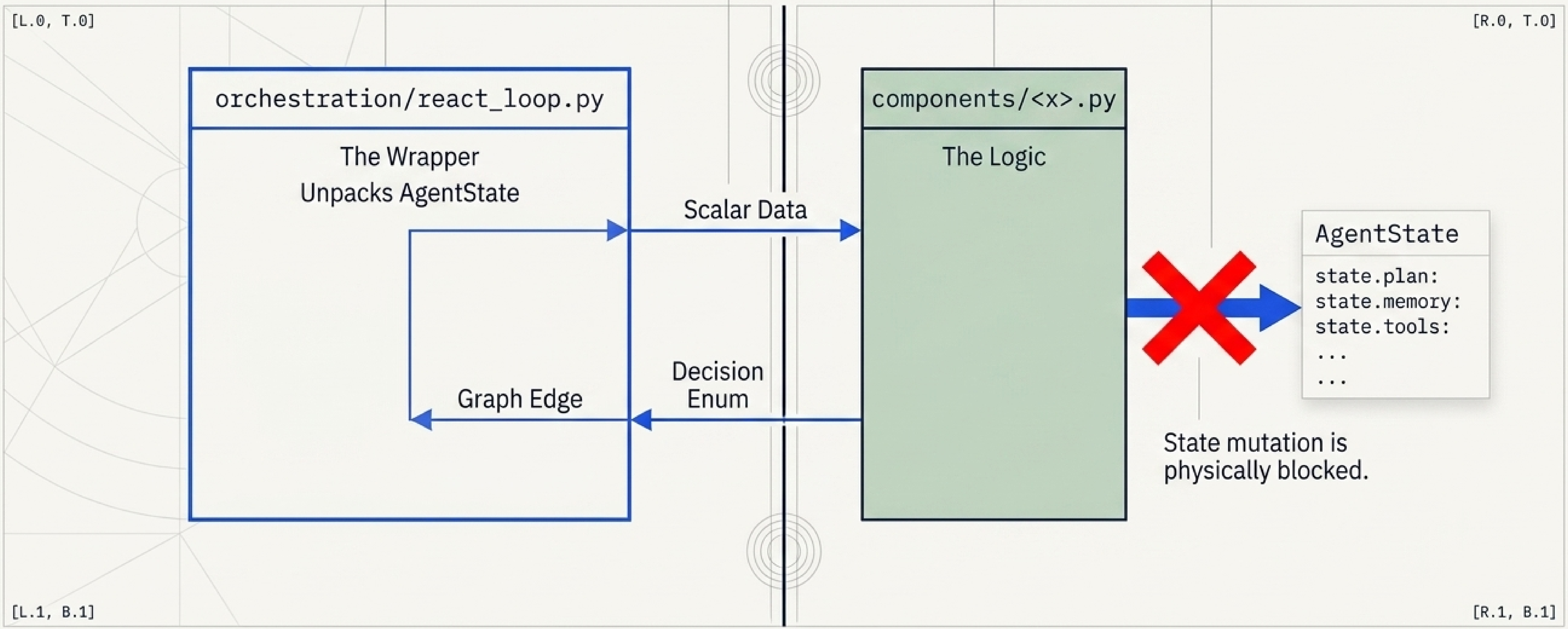
The Onion-Boundary Protocol separates state topology from component logic

OBP-1
Components own computation.

OBP-2
Decisions are returned as pure data enums.

OBP-3
Nodes are thin wrappers that adapt state but contain no logic.

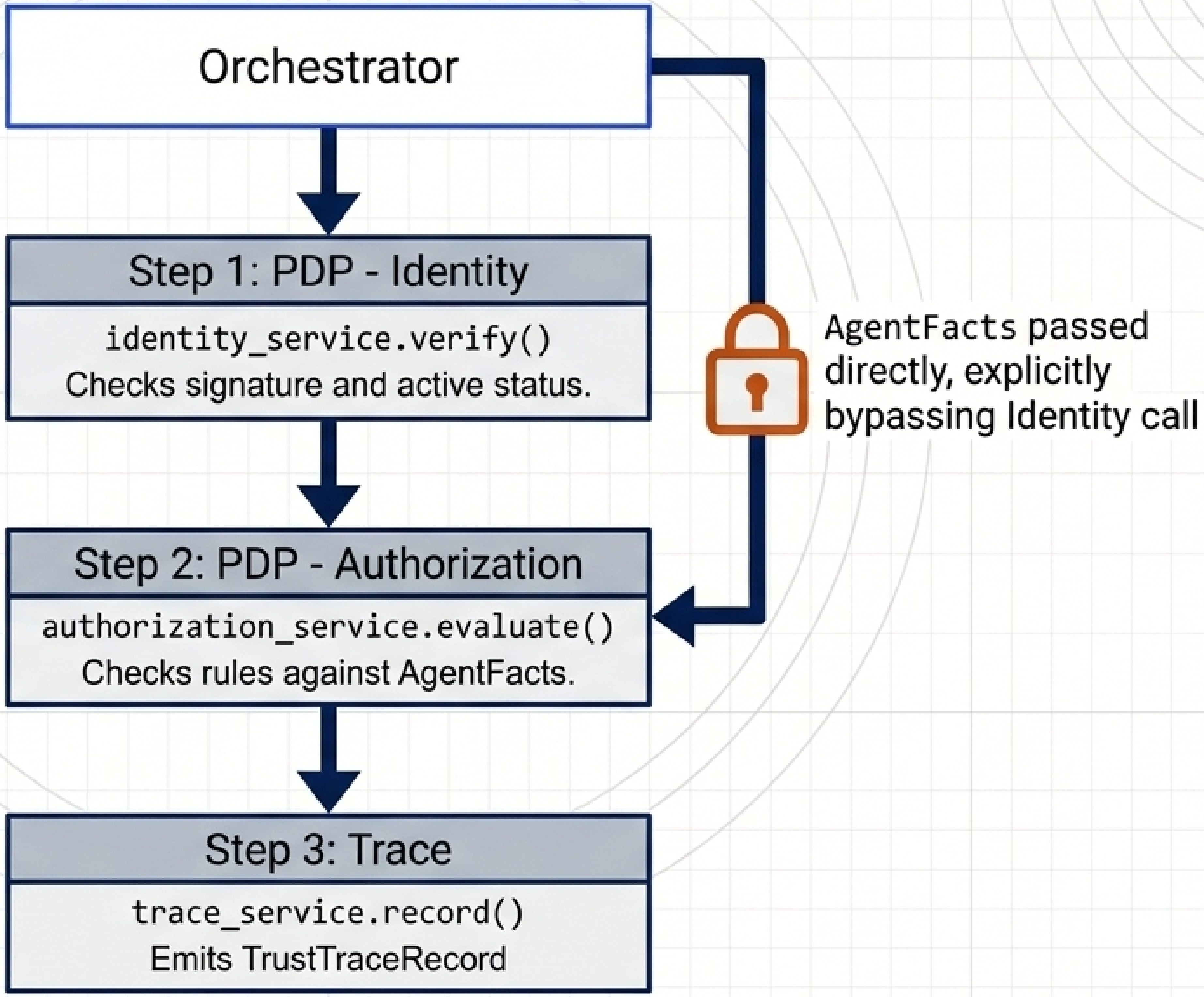
OBP-4
Edges wire the enum to a target without callbacks.



[L.2, T.2]

The Runtime Trust Gate executes a sequential Zero Trust check

The Orchestrator acts as the Policy Enforcement Point (PEP), intercepting every action before it executes.



[R.2, B.2]

Trust is built slowly and lost rapidly through asymmetric scoring

[R.3, T.3] [R.3, B.3]

Access Rings

Ring	Score	Capabilities
Ring 0 (Full Trust)	900-1000	All capabilities.
Ring 1 (Standard)	700-899	Standard capabilities.
Ring 2 (Restricted)	400-699	Read-only.
Ring 3 (Probation)	100-399	Logging only.
Suspended	0-99	Hard gate block.

[R.3, T.3] [R.3, B.3]

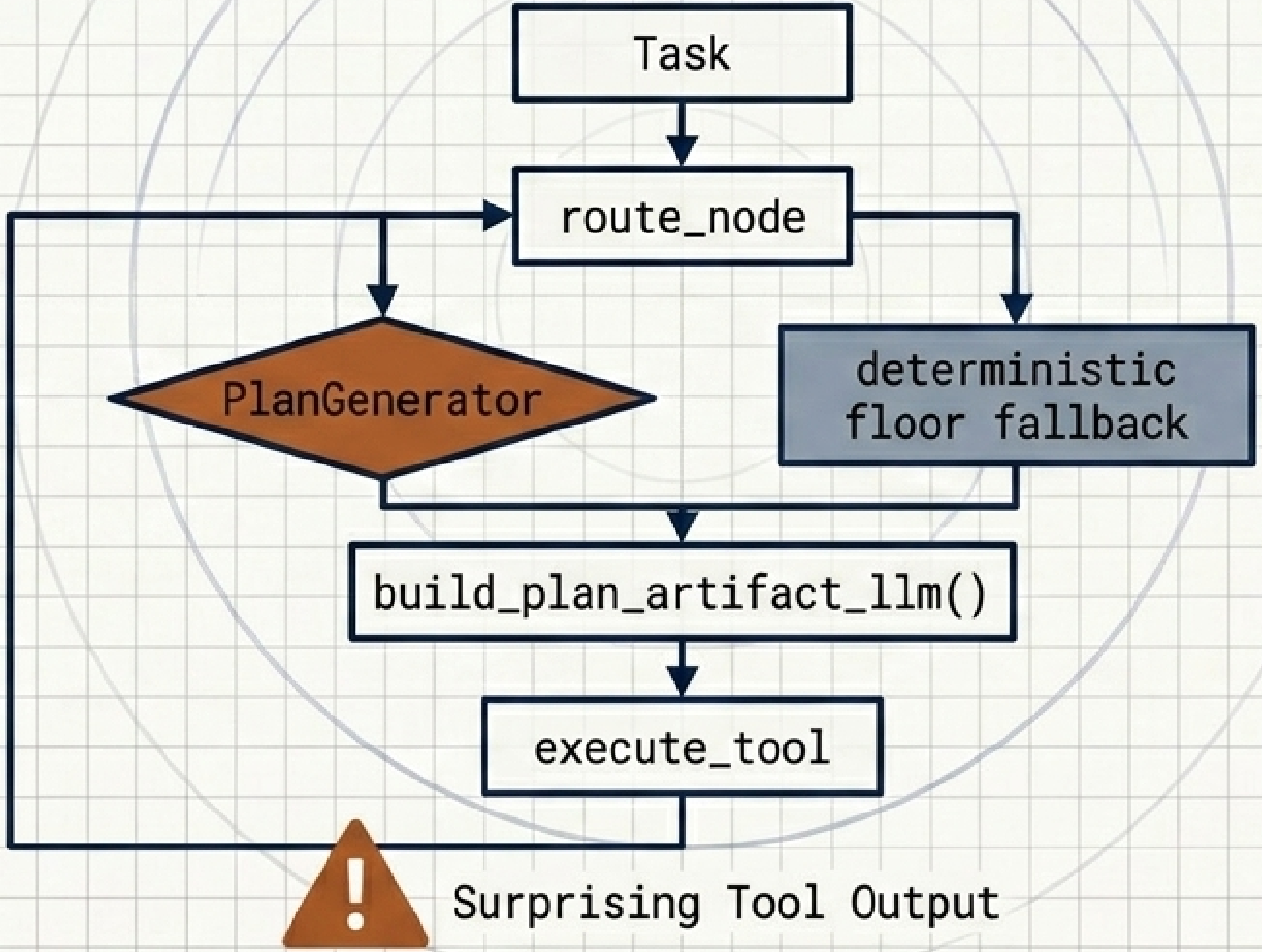
Planning depth and pipeline tiers operate as orthogonal axes

The baseline ReAct spine (T0) can be extended dynamically. Depth controls the step budget and strictness; Tiers control additive pipeline mechanisms.

	L0 (Max 1 step)	L1 (Max 3 steps)	L2 (Max 5 steps)
T1 (LLM Plan Artifact)	Yes	Yes	Yes
T2 (Reflexion Re-entry)	Yes	Yes	Yes
T3 (Supervisor Fan-Out)	Never	If steps >= 2	If steps >= 2

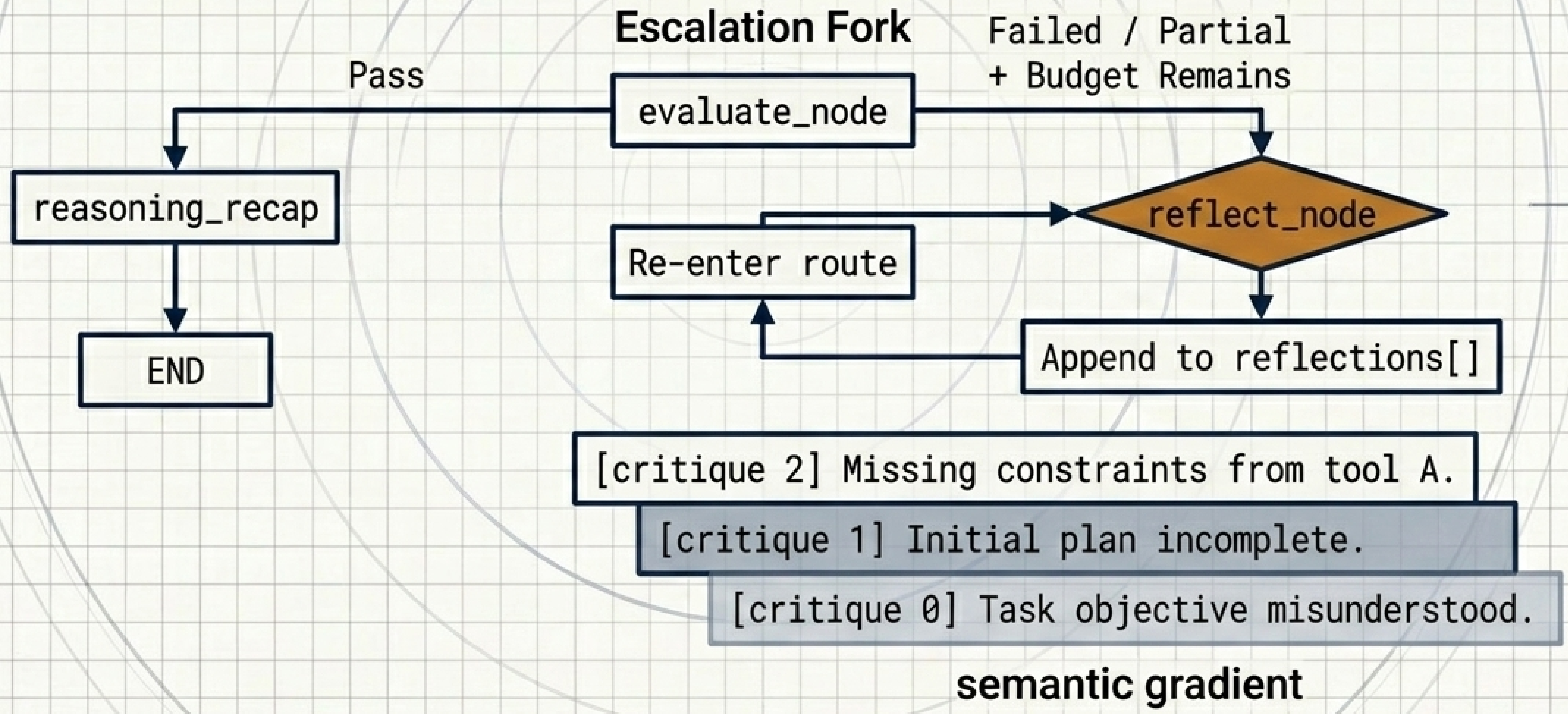
T1 establishes a deterministic plan floor with a dynamic replan edge

When tool outputs deviate from expectations or a plan becomes stale, the system triggers a replan without re-invoking the LLM, relying on a deterministic backstop to prevent brittle plans.



The T2 Reflexion Loop accumulates a semantic gradient of critiques

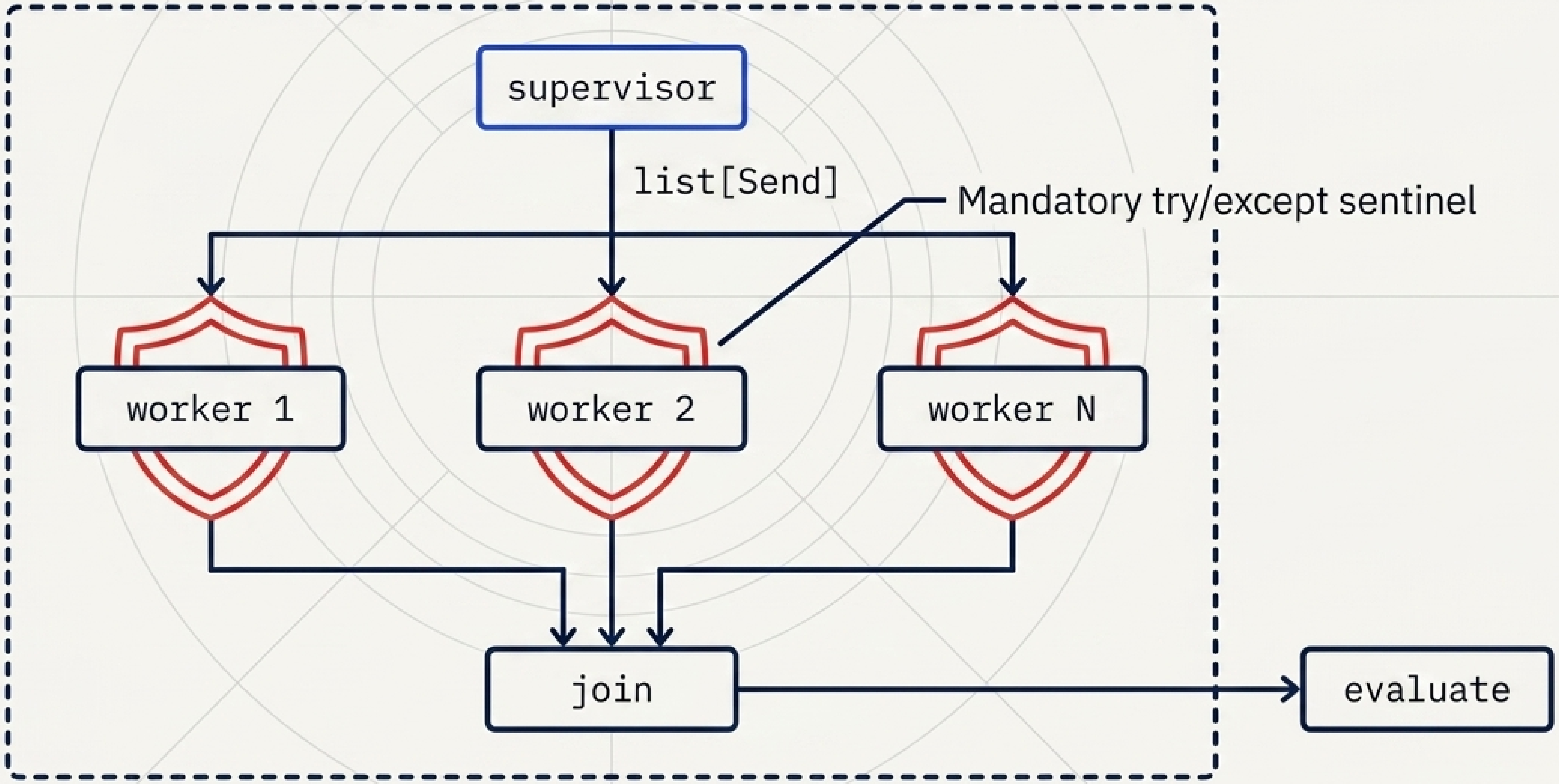
Escalation triggers re-entry. **Critiques** are **strictly append-only** in the state, ensuring prior mistakes survive checkpoint reloads and guide subsequent iterations.



T3 delegates parallel execution with mandatory partial-survival sentinels

Sentinels are not defensive politeness; they are the **only** mechanism preventing a single failed branch from cancelling the entire superstep. The GoalJudge always scores the joined answer, never a fragment.

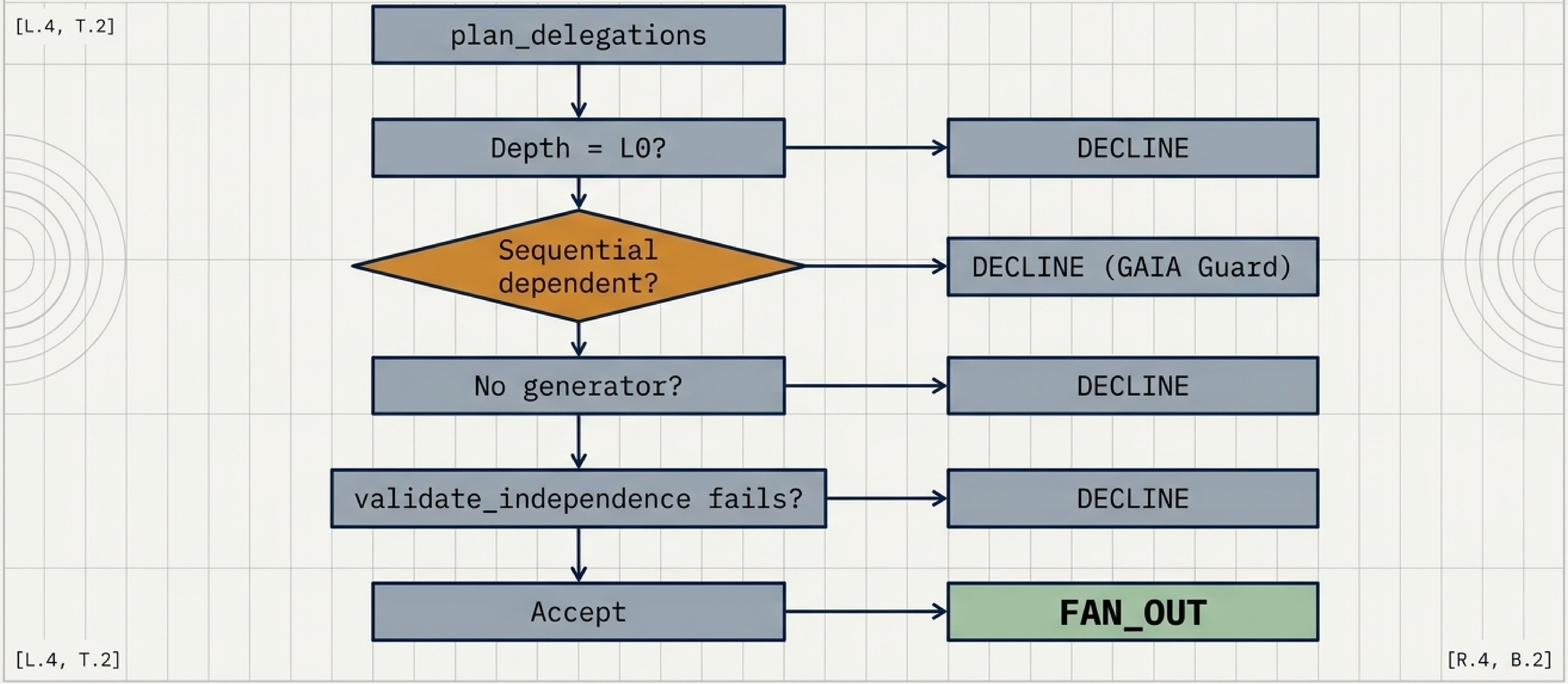
[L.1, T.1]



[R.1, B.1]

The GAIA Guard declines sequential dependencies to prevent parallel corruption

The default return on any path lacking explicit, validated independence is a decline to single-threaded execution. Missing a fan-out is a cheap error; a wrong fan-out is an expensive corruption.



[L.7, T.1]

[R.7, B.1]

Empirical validation confirms zero false fan-outs on coupled structures

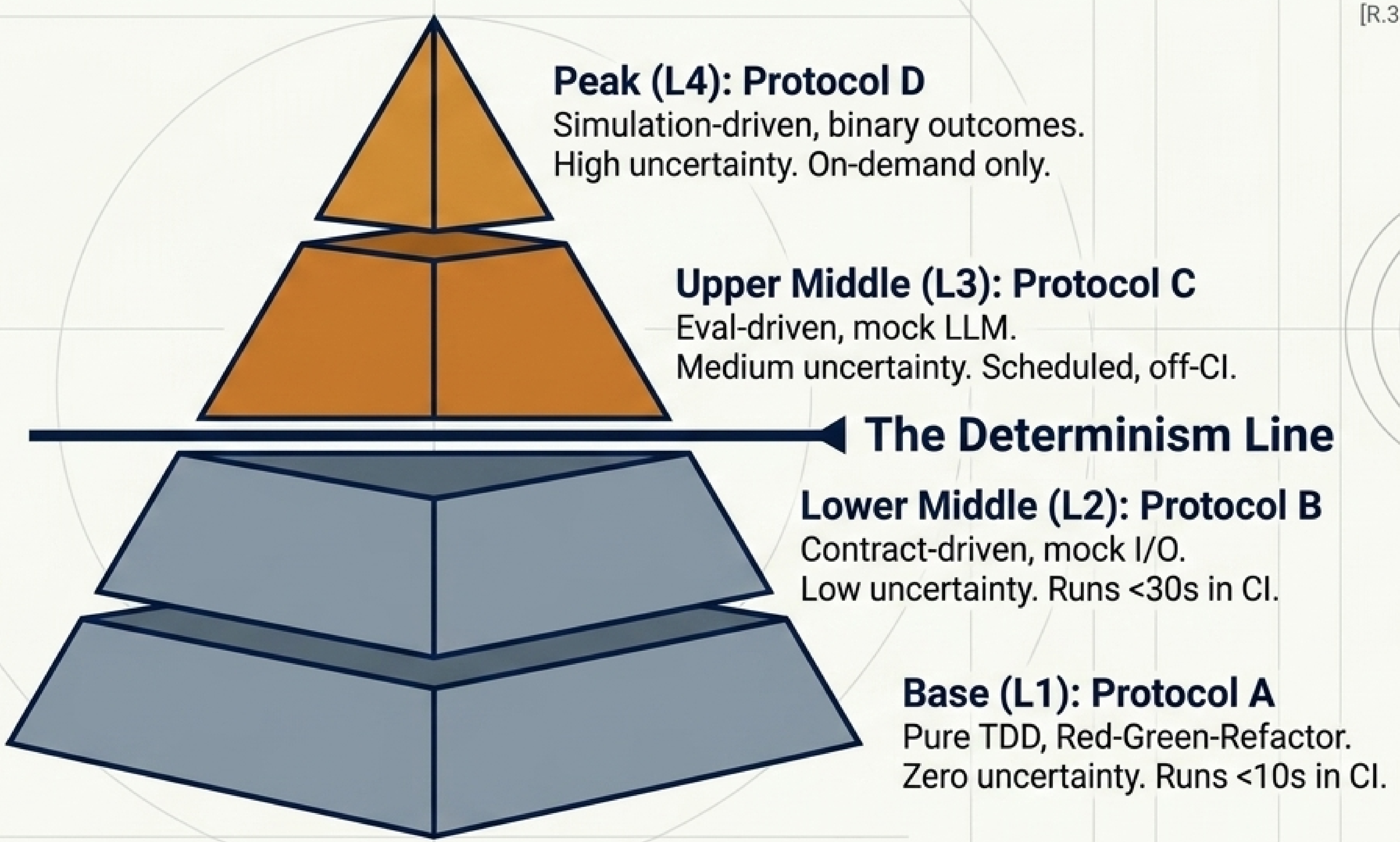
The structure-check overrides model optimism. The system achieves a 1.0 precision rate in declining tasks with shared write targets or hidden sequential dependencies.

Scenario	Description	Outcome
True Independence	3 unrelated region briefs	Fanned out, 3/3 complete
Hidden Coupling	Hotel dates depend on flights	Vetoed by structure validator (Precision 1.0, 0 false fan-outs)
Fault Injection	Timeout token injected	Partial survival measured

[R.7, B.1]

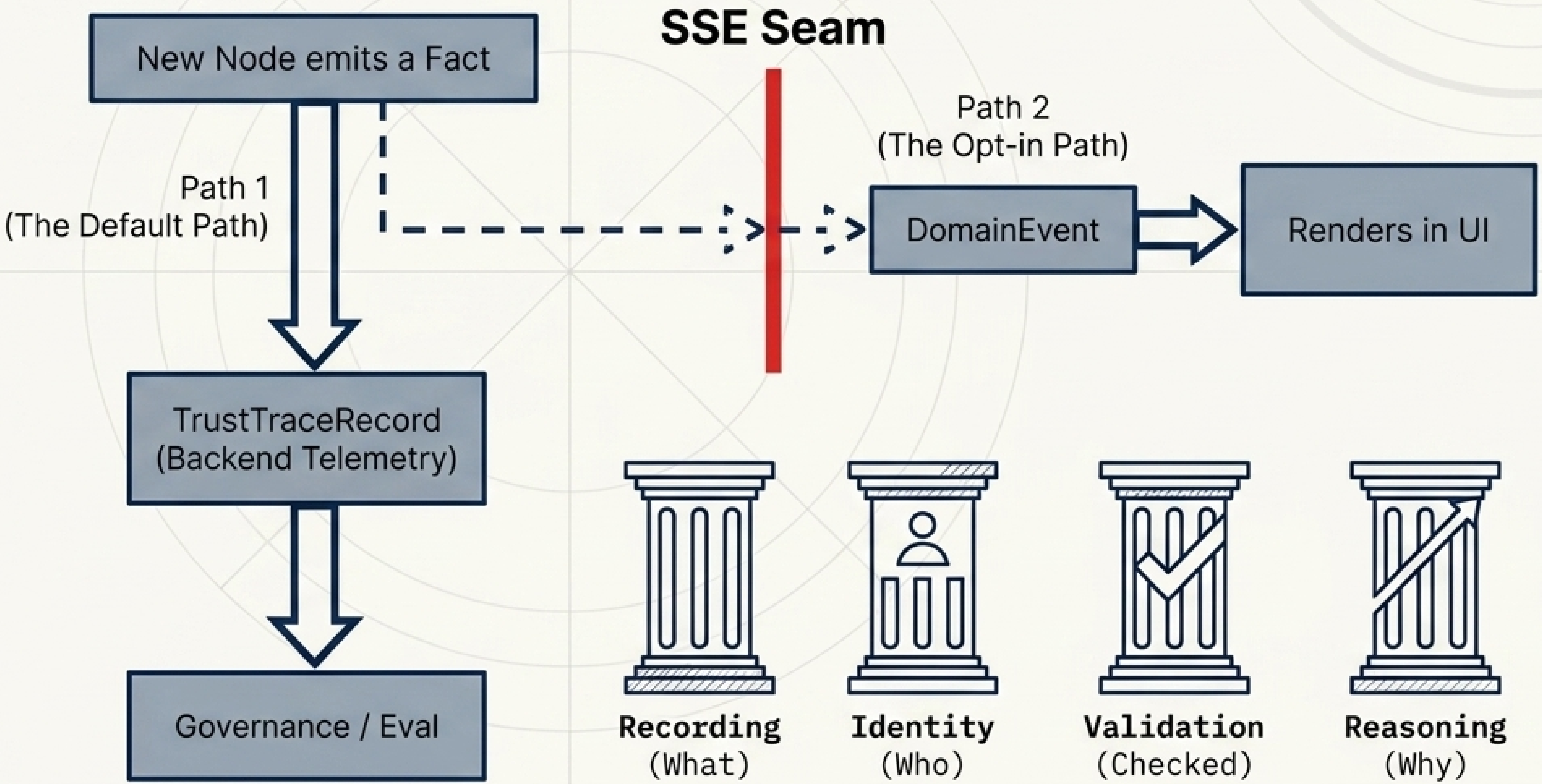
The Agentic Testing Pyramid maps testing protocols to architectural uncertainty

CI only validates deterministic L1/L2 layers. Live LLMs are never executed in CI environments.



Observability defaults to backend telemetry, separated by a strict UI seam

A trace must answer four questions from the trace alone. The governing rule is to curate volume, never truth: every fact must retain exactly one reliable carrier that actually exports.



A secure, governed, self-reflecting, and parallelizing autonomous agent

By completely decoupling structural constraints, runtime tiers, and systemic governance, the architecture allows infinite compositional scaling without compromising trust or determinism.